

1. **Question:** What is the output of the following code?

```
pythonCopy code
```

```
x = 5 y = 2 result = x / y
```

- a) 2.5
- b) 2
- c) 2.0
- d) 2.5 with a remainder

2. **Question:** In Python, which of the following is used for single-line comments?

- a) #
- b) /* */
- c) <!-- -->
- d) --

3. **Question:** What will the following code output?

```
pythonCopy code
```

```
my_list = [1, 2, 3, 4, 5] print(my_list[1:3])
```

- a) [1, 2]
- b) [2, 3]
- c) [1, 3]
- d) [2, 4]

4. **Question:** Which of the following is used to accept user input in Python?

- a) get()
- b) input()
- c) accept()
- d) read()

5. **Question:** What does the len() function do in Python?

- a) Returns the length of a string
- b) Returns the length of a list
- c) Returns the length of a tuple
- d) All of the above

6. **Question:** What is the purpose of the else clause in a try...except block?

- a) It is always executed
- b) It is executed when an exception occurs
- c) It is executed when no exception occurs
- d) It is used to define the exception

7. **Question:** Which of the following is a valid way to open a file in binary mode?

- a) file = open("example.txt", "rb")
- b) file = open("example.txt", "r")
- c) file = open("example.txt", "wb")
- d) file = open("example.txt", "w")

8. **Question:** What is the output of the following code?

pythonCopy code

```
x = True y = False print(not x or y)
```

- a) True
- b) False
- c) Error
- d) None

9. **Question:** Which of the following is used to remove the last element from a list?

- a) `pop()`
- b) `remove()`
- c) `delete()`
- d) `del()`

10. **Question:** What does the `pass` statement do in Python?

- a) Terminates the program
- b) Skips the current block of code
- c) Raises an exception
- d) None of the above

11. **Question:** What does the `str()` function do in Python?

- a) Converts a number to a string
- b) Converts a string to a number
- c) Returns the length of a string
- d) Prints a string

12. **Question:** How can you check if a variable is of a specific type in Python?

- a) `check(type(variable) == desired_type)`
- b) `isinstance(variable, desired_type)`
- c) `type(variable) == desired_type`
- d) `validate(variable, desired_type)`

13. **Question:** What is the purpose of the `continue` statement in Python?

- a) Terminates the loop
- b) Skips the rest of the code in the loop and continues with the next iteration
- c) Jumps to a specific label in the code
- d) Breaks out of the loop

14. **Question:** Which of the following is used to concatenate two lists?

- a) `merge()`
- b) `concat()`
- c) `extend()`
- d) `+`

15. **Question:** What does the `os.path.join()` method do?

- a) Joins two strings
- b) Joins two paths
- c) Joins two lists
- d) None of the above

16. **Question:** In Python, what does the `__init__` method do in a class?

- a) Initializes the class
- b) Initializes the object
- c) Initializes the attributes of the object
- d) Initializes the constructor

17. **Question:** What is the purpose of the `break` statement in Python?

- a) Terminates the program
- b) Skips the current iteration of a loop
- c) Jumps to a specific label in the code
- d) Exits the loop prematurely

18. **Question:** Which of the following is the correct syntax for a list comprehension?

- a) `[x for x in range(10) if x % 2 == 0]`
- b) `{x for x in range(10) if x % 2 == 0}`
- c) `(x for x in range(10) if x % 2 == 0)`
- d) `<x for x in range(10) if x % 2 == 0>`

19. **Question:** How do you import a module named `mymodule` in Python?

- a) `import mymodule`
- b) `include mymodule`
- c) `require mymodule`
- d) `from mymodule import *`

20. **Question:** What does the `random.choice()` function do in Python?

- a) Generates a random number
- b) Selects a random element from a list
- c) Chooses a random character from a string
- d) None of the above